SALVIN
Engineering - Processing - Packaging

LIQUID GLUCOSE MANUFACTURING PLANT

Your Trusted Partner for Consultancy and Turnkey Project Execution

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COMPANY OVERVIEW

Salvin Industries is a leading engineering, technology and consulting conglomerate, working closely with diversified industries spanning across food processing, agriculture, plastics, chemicals, metals healthcare, packaging, Dealer, distributer, super stocks, marketing plan, product analysis and other industries and business sectors offering complete turn-key solutions including advanced engineering, Industry 4.0 solutions, digital transformation and expert technical and management consultancy. With 10+ years of experience and global presence in 8+ countries, Salvin Industries is serving from the most demanding to the most distant customers across the globe. We empower them for rewriting their future with our depths of knowledge, unique ideas, technical expertise, creative solutions and enduring results, all at the most optimal costs, right at their doorsteps.

Armed with the eminence in technological expertise, global capabilities and a process driven approach, we have the ability to take on any challenging project. Crossing the geographical borders & cultures and living up to our core values Excellence, Respect, Care & Joy, we propel innovation to engineer solutions that build & foster equality. Working together as one team with all our business partners & associates, we are transforming industries, fueling economies, developing sustainable solutions and thereby elevating societies, globally. We realize the need to respond to the dynamism & speed with which the world is changing all around us and understand the need to anticipate

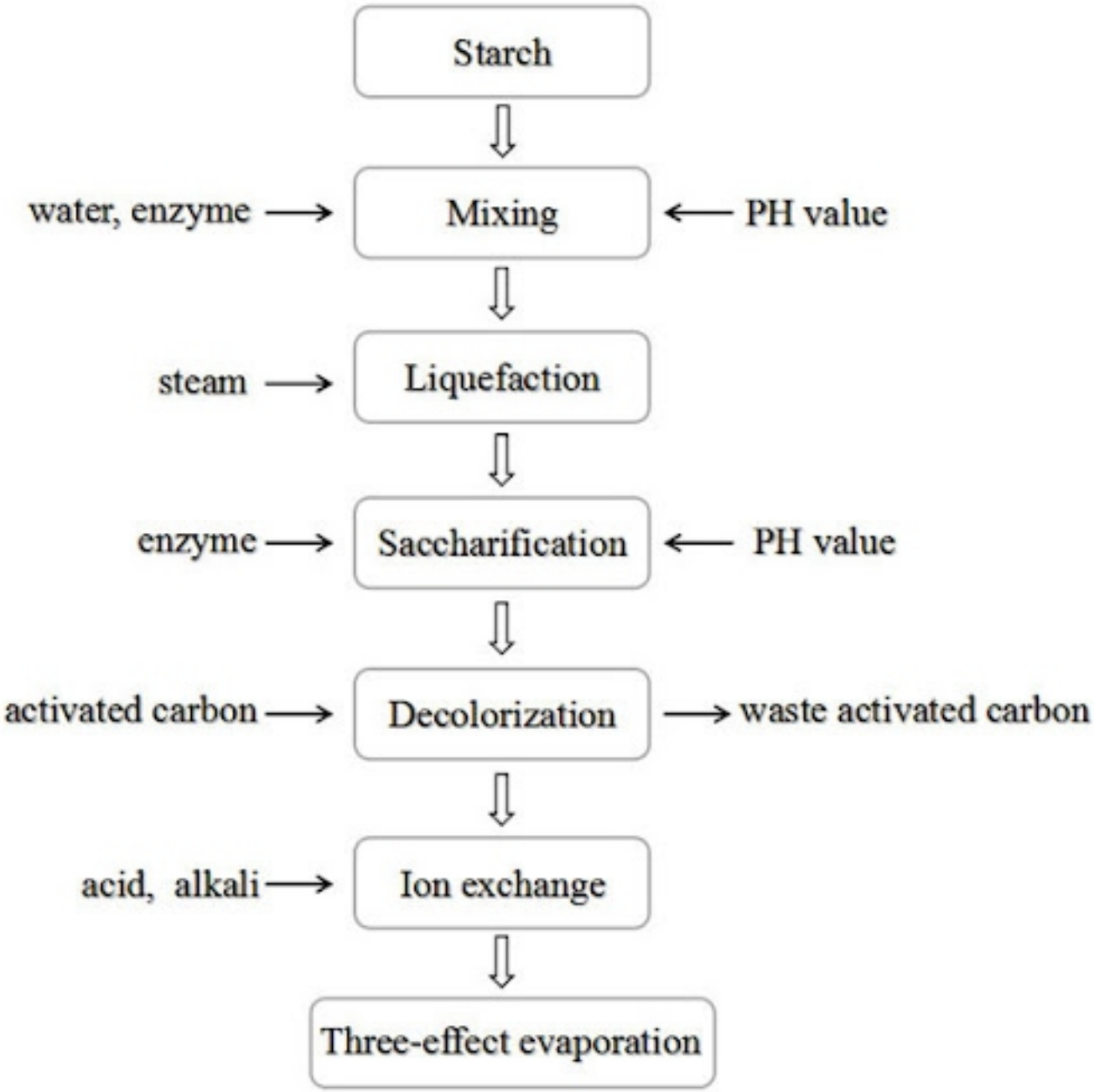
LIQUID GLUCOSE
MANUFACTURING PLANT

Glucose has a famously strong sugar flavor, and it plays an important role in the human body in addition to being very useful in the kitchen. Liquid glucose is a solution of glucose suspended in liquid and is an extremely pure form of glucose which has a concentrated flavor. Glucose is produced commercially via the enzymatic process of Starch and starch is produced from Maize.

Liquid glucose (sp. Gravity 1.43) is a thick, syrupy, colorless to yellowish liquid obtained by the partial hydrolysis of Starch. It contains, water 14.20% dextrose maltose 31-55% ash, 0.5-0.8% and per obtain 0.02%.

starch	74%
Protein	9.2%
fat	1.6%
Fibers	1.2%
Moisture	12%
others	2%

FLOW CHART LIQUID GLUCOSE MANUFACTURING PLANT



LIQUID GLUCOSE MANUFACTURING PLANT

THE PROCESS

Starch Mixing

To create a homogeneous starch slurry with optimal conditions for enzymatic action.

Starch powder (e.g., corn, cassava) is mixed with water to achieve a specific dry solid (DS) concentration, typically 28-32% . The pH and temperature of the slurry are precisely adjusted at this stage to the optimal range for the enzyme that will be added in the next step. Food-grade acids or alkalis (like soda ash or sulfuric acid) are used for pH adjustment .

Liquefaction

To break down long starch chains into shorter dextrins, reducing viscosity and preparing the substrate for saccharification.

This is a high-temperature enzymatic process. The starch slurry is dosed with a heat-stable alpha-amylase enzyme. The mixture is then instantly heated to high temperatures (105-110°C) using a jet cooker to gelatinize the starch and allow the enzyme to work quickly. It is then held in a series of liquefaction columns or tanks to complete the breakdown to dextrins. This step is critical for achieving the right Dextrose Equivalent (DE) for the next stage

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THE PROCESS

Saccharification

To convert the dextrins from liquefaction into individual glucose molecules.

The liquefied starch is cooled to around 55-60°C and the pH is adjusted to an optimal level (typically 4.0-4.5). Glucoamylase enzyme is then added. This reaction takes place in large, agitated vessels over a period of 24-72 hours. The progress is monitored by tracking the DE value, which will rise to 95-98% for high-conversion syrups .

Decolorization

To remove color bodies, residual proteins, and other impurities to achieve a clear, high-quality syrup.

The glucose syrup is heated and mixed with activated carbon. The carbon particles adsorb organic impurities that cause color and off-flavors. After a controlled contact time, the carbon, now laden with impurities, must be completely removed from the syrup. This is typically done using a filter press to ensure a brilliant, clear filtrate.

LIQUID GLUCOSE MANUFACTURING PLANT

THE PROCESS

Ion Exchange

To demineralize the syrup by removing ionic impurities like salts, amino acids, and other minerals, producing a high-purity, food/pharma-grade product.

The decolorized syrup is passed through a series of columns containing ion exchange resins. This usually involves a cation column (to remove positive ions) followed by an anion column (to remove negative ions). Modern plants use automated systems with sensors to monitor the process and trigger automatic regeneration of the resins when they are exhausted, ensuring consistent quality .

Three-Effect Evaporation

To concentrate the purified, dilute glucose syrup to the final commercial solids concentration (typically 70-85% DS).

This is an energy-efficient method of concentration. The syrup is boiled under vacuum in a series of three evaporators (effects). The vacuum lowers the boiling point, protecting the syrup from heat damage. The key principle is that the vapor (steam) generated from boiling the syrup in the first effect is used as the heating medium for the second effect. This "reuses" the energy, making the process much more efficient than single-effect evaporation .

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TOGETHER

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